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The Cosmic Microwave Background

Introduction

The early universe was a hot place; all the atoms you're familiar with were ionized. Photons, those little particles of light, constantly scattered off of the roaming free charges. Assuming you could survive being there, you wouldn't be able to see anything! As the universe expanded, it cooled; a phase transition occurred when its temperature was so low that electrons could start binding with protons. During this *recombination*, the neutralization of the cosmic plasma resulted in the very fast growth of the photons' mean free path. When the path became larger than the Hubble radius, the universe became transparent. Photons scattered one last time, carrying the information coming from that scattering in their physical attributes, and began their long journeys through the ever emptier and cooler cosmos.

Along their myriad journeys, some of them were absorbed. Some of them encountered the gravitational wells of giant clusters of matter, and had to climb in and out of them. A small fraction of them encountered a stray free charge, scattered and kept on flying. An even smaller fraction have done that twice in the history of all there is! For the most part, however, their journeys have been rather uneventful. In fact, there are on average about 400 of these photons in every cubic centimeter of space, each carrying information about that last scattering that occurred so very long ago. By just standing there, you are continuously absorbing light that has not interacted with matter for almost 14 billion years!

Measuring these incoming photons from every direction in the sky, we get a picture of the universe from then. That's what people will tell you, but what does it mean? Well, a particularly useful quantity one can measure about these photons is their frequency. For a fixed direction in the sky, count the incoming photons and jot down the frequency of each one. Once enough of them have been gathered to satisfy the Sultan of statistics, examine the resulting distribution, calling $dn = dn(\nu)$ the number of photons between frequencies ν and $\nu + d\nu$. As it turns out, the distribution one finds happens to be that of a blackbody spectrum:

$$\frac{dn}{d\nu} = \frac{8\pi\nu^2}{c^3} \frac{1}{e^{\frac{h\nu}{kT}} - 1} \quad (1)$$

In fact, it is the most exquisitely accurate blackbody spectrum ever found. So, now one has a blackbody spectrum associated with every point in the sky. But, are they the same spectrum, or are they different?

The only free parameter in equation 1 is the temperature, T . So for direction $\hat{n}$ take the corresponding blackbody and extract the temperature $T = T(\hat{n})$. Doing this for every point in the sky results in a temperature map: the Cosmic Microwave Background (CMB). The map made from the Planck Satellite data is shown in figure 1. This name may give you the impression that the CMB is far away, out there on the sky, in the background. But remember, these are the 400ish photons in every cubic centimeter of space. You're swimming in this Cosmic Microwave Bath whenever you have the pleasure of seeing the sky!

Ok, ok, enough waxing poetic. What do we learn from this map? Well, it turns out that all those blackbodies we collected are almost exactly the same, but not quite. The temperature in each direction will be ever so slightly different than the directions near it. This is a profound observation, and leads us to two very interesting questions: (1) Why are all the directions nearly the same temperature? (2) Why is each direction slightly different than its neighbors? These may sound like the same question, but they are not. The first is a question about commonality - all the photons in the CMB have experienced something identical: the expansion of the universe. The second is about uniqueness - how photons in different parts of the universe experienced slightly different things: the peculiarities of recombination.

So, exactly how similar are all the blackbodies? Integrating over all solid angles gives the average value of the temperature map:

$$\bar{T} = \frac{1}{4\pi} \int d\Omega T(\hat{n}). \quad (2)$$

When this is done one finds $\bar{T} = 2.7255\text{K}$. To be free of the arbitrariness inherent in the choice of a temperature scale, define

$$\delta_T(\hat{n}) = \frac{T(\hat{n}) - \bar{T}}{\bar{T}} \quad (3)$$

δ_T is the fluctuation map of the CMB. For every direction in the sky the fluctuations are $|\delta_T| < 10^{-3}$; incredibly tiny! In figure 2 the histogram of δ_T is plotted with a Gaussian overlaid (in a logarithmic scale a Gaussian is an downward parabola). We say that the fluctuations form a Gaussian random field. You may notice that there are deviations in the tails of the Gaussian, starting around 4σ . If you were really observant, you might also wonder if the differences are asymmetric. We will have more to say about non-Gaussianity later. I should also note that the language is being a bit abused here.

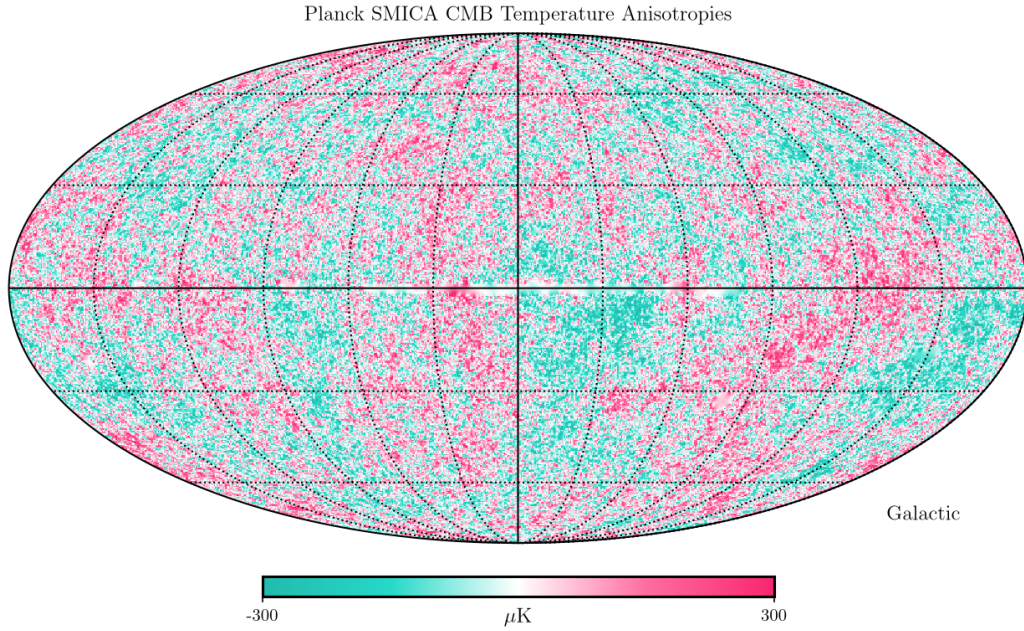


Figure 1: The map of the CMB temperature anisotropies. The image shows the whole sky in what is known as a Mollweide projection. This is in Galactic coordinates, so the center of the plot is the center of the Milky Way galaxy. Both the monopole and the dipole have been removed from this data.

Why do $\bar{T}$ and δ_T have the values that they do? As mentioned earlier, $\bar{T}$ comes from the expansion of the universe. Thinking in temperature is ok, but its better to associate an energy scale with the CMB, $E_{CMB} = k_B \bar{T} \approx 3.76 \times 10^{-23} \text{J}$ or $2.35 \times 10^{-4} \text{eV}$. Recall that the reason recombination happened was that the universe was cooling. It's been cooling every since, the CMB acting like the gas in an expanding piston. The ionization energy of Hydrogen is $E = 13.6 \text{eV}$, but, because the matter density was much higher during recombination, the energy scale of the CMB was closer to $E_{Rec} \approx 0.26 \text{eV}$ (this would have been sufficient to ionize enough of the plasma to make the universe opaque). So the photons were $E_{Rec}/E_{Rec} \approx 1,100$ times more energetic when they scattered that last time. They lost energy because the expansion of the universe stretched their wavelengths and $E = hc/\lambda$ for a photon. So the reason that $\bar{T}$ has the value it does is because the universe has expanded by a factor of $\sim 1,100$ since recombination occurred.

The picture of the universe that the CMB reveals is one where the cosmos was three orders of magnitude tinier than it is today! An infant universe. What about those temperature fluctuations? What can they tell us? It turns out to be A LOT. But to get there, we have to bring some order to the fluctuations. Statistical tools were constructed to analyze fluctuating quantities, so to proceed we will have to done the hats of statisticians. We already started by plotting the histogram of temperature fluctuations in figure 2. There, a Gaussian profile fits the data beautifully, with a best fit standard deviation of $\sigma = 3.98 \times 10^{-5}$. The variance (σ^2) tells us the scale of the variation relative to the mean of each direction in the sky. Pondering such things leads one to wonder about

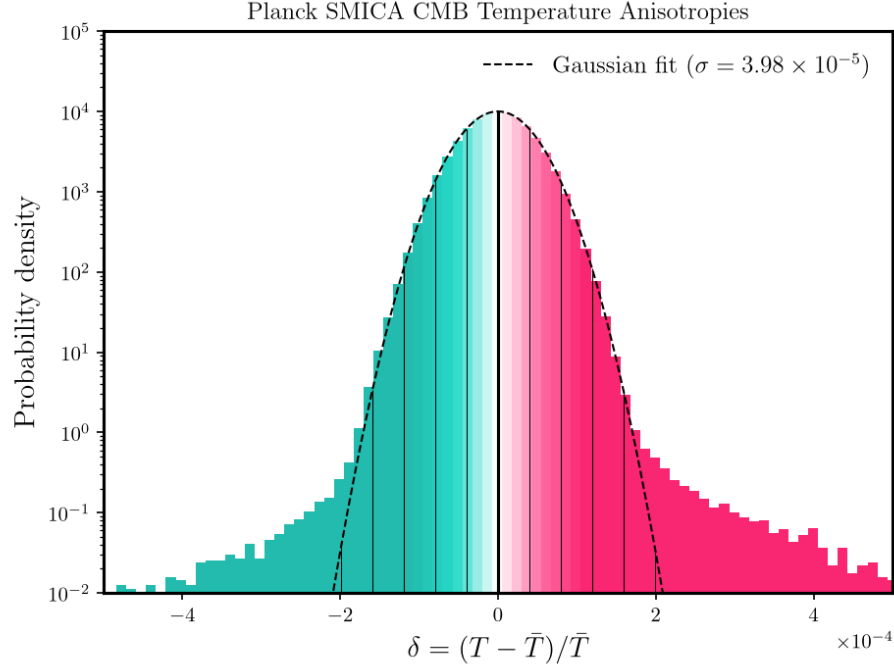


Figure 2: The histogram of temperatures. The data fits a Gaussian remarkable well, with deviations occurring at the 4σ level.

the co-variation between different directions in the sky.

To calculate the covariance of the fluctuations in the sky one needs a notion of distance between directions? Since the map is on a sphere, the natural "distance" between points is their angular separation. Pick an angle, then point in two directions, $\hat{n}$ and $\hat{n}'$, making sure that they are separated by your chosen angle, $\hat{n} \cdot \hat{n}' = \cos \theta$. Take the product of the variations in those directions, $\delta_T(\hat{n})\delta_T(\hat{n}')$. Then take the average over all such directions:

$$C(\theta) = \frac{1}{4\pi} \int d\Omega \int d\Omega' \delta_T(\hat{n})\delta_T(\hat{n}') \delta(\hat{n} \cdot \hat{n}' - \cos \theta) \quad (4)$$

this is the famous 2-point, or angular, correlation function. When we let $\theta = 0$, so that the two directions in the sky are actually the same, we recover the variance,

$$C(0) = \frac{1}{4\pi} \int d\Omega \delta_T(\hat{n})^2 = \sigma^2. \quad (5)$$

The Monopole

A quick calculation will show that $a_{00} = 0$ in the expansion. This is physically obvious since we are examining the deviation from the mean, and the spherical harmonic $Y_{00} = \frac{1}{\sqrt{4\pi}}$ is a constant. We need not include it in the expansion.

The Dipole

Consider an observer moving relative to the CMB rest frame with some velocity $\vec{v}$. Due to their motion they will measure Doppler shifted CMB photons, changing the reconstruction of their temperature map as compared to an observer in the CMB rest frame via:

$$T_{\text{obs}}(\hat{n}) = \frac{1 + \frac{\hat{n} \cdot \vec{v}}{c}}{\sqrt{1 - \frac{v^2}{c^2}}} T(\hat{n}) \quad (6)$$

This Doppler shift will introduce mixing between the actual expansion coefficients due to the $\cos \theta$ term coming from the dot product with the observer's velocity wrt the CMB. One can show that this mixing is strongest for the dipole, where the monopole of the actual expansion contaminates the observed one. This means that the relative motion of the galaxy to the cosmic microwave background contributes most of the dipole measurement. Disentangling this contamination from the actual dipole is difficult, so we will ignore the dipole in the spherical expansion.

Gaussian Approximation

The Gaussian approximation posits that we independently draw the expansion coefficients for each ℓ from the same Gaussian distribution, $a_{\ell m} \sim \mathcal{N}(0, C_\ell^{\frac{1}{2}})$:

$$\langle a_{\ell m} \rangle = 0 \quad (7)$$

$$\langle a_{\ell m}^* a_{\ell' m'} \rangle = \delta_{\ell \ell'} \delta_{m m'} C_\ell \quad (8)$$

Here we have introduced the C_ℓ 's as the variances of these distributions. Under rotations the $a_{\ell m}$'s only mix with themselves for fixed ℓ , this means that the expansion can have a different axis for each ℓ . The choice of axis will have no effect on the C_ℓ 's, making them a rotationally invariant description of the expansion. A quantitative model of the cosmos should make predictions about their values.

Note: the bracket averages above run with respect to the distribution we draw the coefficients from:

$$\langle f(a_{\ell m}) \rangle = \int \mathcal{D}\{a_{\ell m}\} f(a_{\ell m}) \rho(\{a_{\ell m}\}) \quad (9)$$

where the measure is over all coefficients.

Estimating the Variance

We introduce the estimator of the variance as

$$\hat{C}_\ell = \frac{1}{2\ell + 1} \sum_m |a_{\ell m}|^2 \quad (10)$$

Imagine observers sprinkled throughout the universe. Each of them will see a different CMB, and therefore measure different values of $\hat{C}_\ell$. If, through some feat of unforeseen physics, these observers could collect all their values, what sort of statistics would they discover. First they would find that

the estimator is unbiased, $\langle \hat{C}_\ell \rangle = C_\ell$. The mean of all their measurements would be the actual C_ℓ that the universe realized.

To find the higher moments (cumulants) of the estimator distribution, note that $\hat{C}_\ell$ is just the average of the squares of a bunch of Gaussian variables. We encountered the resulting probability density earlier, and now state the result:

$$\rho(\hat{C}_\ell | C_\ell) = \frac{(\ell + \frac{1}{2})^{\ell + \frac{1}{2}}}{\Gamma(\ell + \frac{1}{2})} \frac{\hat{C}_\ell^{\ell - \frac{1}{2}}}{C_\ell^{\ell + \frac{1}{2}}} e^{-(\ell + \frac{1}{2}) \frac{\hat{C}_\ell}{C_\ell}} \quad (11)$$

One can quickly verify the earlier statement that the estimator is unbiased (exercise). Furthermore, the variance in the estimator is:

$$\langle \hat{C}_\ell^2 \rangle = \frac{C_\ell^2}{\ell + \frac{1}{2}} \quad (12)$$

We see that as $\ell \rightarrow \infty$, the variance vanishes. Observers can be very confident that high ℓ estimators are nearly identical to the actual values. For small ℓ , however, they cannot be so confident. The reason for this lies in the number of $a_{\ell m}$'s needed to make each estimator. For large ℓ , there are many, and by the central limit theorem, their average becomes peaked at the true average. On the other end this is not the case. Cosmologists have dubbed the uncertainty associated with the low ℓ estimators *cosmic variance*.

The Probability of a Cosmos

The Cosmos will have some value for the C_ℓ 's. An observer in the Cosmos will have estimators, $\hat{C}_\ell$'s, once their technology is sufficient to measure the CMB. Given an observer's estimators, what is the probability that the Cosmos has realized a particular set of the C_ℓ 's? This is a very important question for the observer, since they wish to know what Universe they actually reside in.

We can frame this question in Bayesian analysis. The set of actualized values that the world realized forms the model, $M = \{C_\ell\}$. Meanwhile, the set of observed estimators forms the data, $D = \{\hat{C}_\ell\}$. The strength of belief in a particular model that the observers hold is then the posterior after they make the measurements:

$$p(M|D) = \mathcal{L}(D, M)p(M) \quad (13)$$

$$\mathcal{L}(D, M) = \frac{p(D|M)}{p(D)} \quad (14)$$

Here $p(M|D)$ is the *posterior*, $\mathcal{L}(D, M)$ is the *likelihood*, and $p(M)$ is the *prior*. The relationship between these three reads: the likelihood of the data updates the prior belief that the observer holds in the model, yielding an informed posterior belief. The observer should believe most strongly in the model that maximizes the posterior. As an added feature, Bayesian analysis also tells us exactly how much less the observer should believe nearby models.

One should also immediately ask about the prior. What distribution should observers choose for the C_ℓ 's prior to making any measurements? Here the Maximum Entropy principle once again saves us: we should choose the distribution that stays maximally ignorant under the known constraints.

Since no measurements constrain the prior, MaxEnt demands that it be uniform. The observer should believe no model any more strongly than any other when no data constrains belief.

With regards to the likelihood, we already have the numerator from the previous section. The denominator does not depend on the model parameters, so we treat it as a constant. The posterior density reads

$$\rho(M|D) \propto \prod_{\ell} \left(\frac{\hat{C}_{\ell}}{C_{\ell}} \right)^{\ell + \frac{1}{2}} e^{-(\ell + \frac{1}{2}) \frac{\hat{C}_{\ell}}{C_{\ell}}} \quad (15)$$

$$\propto e^{-\frac{1}{2} \chi_{\text{eff}}^2} \quad (16)$$

$$\chi_{\text{eff}}^2 = \sum_{\ell} (2\ell + 1) \left(\frac{\hat{C}_{\ell}}{C_{\ell}} + \ln \frac{C_{\ell}}{\hat{C}_{\ell}} - 1 \right) \quad (17)$$

where we have introduced an effective chi-squared function, χ_{eff}^2 . Furthermore, we have normalized it so that the model which maximizes the posterior gives a $\chi_{\text{eff}}^2 = 0$. The sum is over all the ℓ values that the observations are over. Minimizing this function (maximizing the posterior) with respect to the model parameters gives us:

$$\nabla p(M|D) = 0 \Rightarrow C_{\ell} = \hat{C}_{\ell} \quad (18)$$

The model that maximizes the posterior is the one in which the parameters the universe realized equal the estimators.

Physical Cosmology

Up until now we have been discussing the observational side of cosmology: the CMB. We know why the CMB should be there, and under the Gaussian assumption can judge the observed angular power spectrum with any particular model. The question is, what is the model that describes the statistics of the CMB? To answer this question we need to delve into the realm of physical cosmology to see how the physics of the early universe imprinted itself on the temperature anisotropies of the photons making up the CMB.

The FRW metric

On large scales we hypothesize that the Universe is isotropic and homogenous. As it turns out, observational data of galaxy cluster distributions supports this assumption. Isotropy means that the universe looks the same in every direction; there is no special direction that breaks this symmetry. Homogeneity means that no point in the universe is any different than any other. Now, of course, on small scales, both these assumptions no longer hold. This is because small scales decouple from cosmic evolution in a process known as hierarchical structure formation, which we shall examine later.

On large scales, however, isotropy and homogeneity hold, so we would like to have an expression for the possible models of spacetime that satisfy these two principles. There is an elegant geometric

way to derive this class which we shall now do. The surface of a sphere satisfies both homogeneity and isotropy, so consider a $D + 1$ dimensional Euclidean space, and let us embed a D -sphere in it,

$$S^D(R) = \{(x_1, \dots, x_D, w) \in \mathbb{R}^{D+1} \mid x_1^2 + \dots + x_D^2 + w^2 = R^2, R \in \mathbb{R}^+\} \quad (19)$$

where R is the radius of the sphere. Next we will look at what the induced metric on the sphere looks like that it inherits from the embedding space. We do this by treating one of the coordinates as a function of the others, $w = w(x_1, \dots, x_D)$. The induced line element reads

$$\begin{aligned} dl^2 &= dx_1^2 + \dots + dx_D^2 + dw^2 \\ &= dx_1^2 + \dots + dx_D^2 + \frac{(x_1 dx_1 + \dots + x_D dx_D)^2}{R^2 - x_1^2 - \dots - x_D^2} \\ &= \frac{dr^2}{1 - \frac{r^2}{R^2}} + r^2 d\Omega^2 \end{aligned} \quad (20)$$

where in the last line we have used spherical coordinates from the D remaining coordinates.

Functions on a Sphere

For many domains we know that there are families of functions that act as the atoms from which one can build other functions. In this section we'll show how there exist such functions on the sphere, the bane of starting physics students, the spherical harmonics!

Consider the unit sphere. Choose an axis and construct a coordinate system from it, $\theta \in [0, \pi]$ and $\phi \in [-\pi, \pi]$. The Laplacian operator with respect to that axis is

$$\nabla^2 = \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \sin \theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2}. \quad (21)$$

The spherical harmonics are the eigenfunctions of the Laplacian, $\nabla^2 Y_\ell^m = -\ell(\ell + 1)Y_\ell^m$, where $\ell \in \mathbb{Z}^+$ and $m \in \{-\ell, \dots, \ell\}$. Since there are $2\ell + 1$ different m 's for each ℓ , the corresponding eigenspaces have the same degeneracy. They satisfy several important properties that, if memorized, will give you the power to manipulate them with ease.

First they satisfy an orthonormality relation

$$\int d\Omega Y_\ell^m(\hat{n})^* Y_{\ell'}^{m'}(\hat{n}) = \delta_{\ell\ell'} \delta_{mm'}. \quad (22)$$

Next, on each eigenspace the spherical harmonics satisfy an addition property

$$\sum_{m=-\ell}^{\ell} Y_\ell^m(\hat{n})^* Y_\ell^m(\hat{n}') = \frac{2\ell + 1}{4\pi} P_\ell(\hat{n} \cdot \hat{n}') \quad (23)$$

where the P_ℓ are the Legendre polynomials. The complex conjugate of a harmonic is

$$Y_\ell^m(\hat{n})^* = (-1)^m Y_\ell^{-m}(\hat{n}). \quad (24)$$

Lastly and most importantly, the harmonics are complete. Meaning that for most every scalar function on the sphere $f : S^2 \rightarrow \mathbb{C}$, it is possible to write it in terms of them,

$$f(\hat{n}) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} a_{\ell m} Y_{\ell}^m(\hat{n}) \quad \text{where} \quad a_{\ell m} = \int d\Omega Y_{\ell}^m(\hat{n})^* f(\hat{n}). \quad (25)$$

That's all well and good, but recall that you had to choose an axis at the start. What if you had chosen another axis to do the spherical expansion in equation 25? Remarkably, the eigenspaces (fixed ℓ) are invariant under rotation, $\hat{n} \mapsto \hat{n}' = R\hat{n}$.

$$Y_{\ell m}(\hat{n}') = \sum_{m'=-\ell}^{\ell} d_{mm'}^{(\ell)}(R) Y_{\ell m'}(\hat{n}) \quad (26)$$

$$\sum_{m'=-\ell}^{\ell} |d_{mm'}^{(\ell)}(R)|^2 = 1 \quad (27)$$

where the d 's form a representation of the rotation operator. Since the temperature transforms as a scalar under rotations, $\delta'_T(R(\hat{n})) = \delta_T(\hat{n})$, this implies that $a_{\ell m}$ transforms as a linear combination of the other $a_{\ell m'}$'s.

Probability Theory

One of the most profound results in probability theory is the central limit theorem. Put simply, a Gaussian distribution approximates the statistics of sums of random variables (rv). The approximation becomes better and better the more rv's we use in the sum. This holds irrespective of the distribution we draw the rv's from.

Our goal will be to look at how good this approximation is. To that end we will develop the *Edgeworth expansion* to see how to expand a limiting distribution around a Gaussian. We will find that the moments of the underlying distribution play a role in the error terms. In this way we will discover a natural way to quantify *non-gaussianity* in a distribution.

The Gaussian Distribution

No discussion of physics is ever devoid of Gaussians, so let's start there. A Gaussian distribution is the maximum entropy distribution with constraints on its mean and variance. More precisely, if we have that

$$\begin{aligned} \langle X \rangle &= \mu \\ \langle X^2 \rangle &= \sigma^2 \end{aligned} \quad (28)$$

then the MaxEnt distribution density reads:

$$\rho_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}} \quad (29)$$

We then say X follows a normal distribution and write $X \sim \mathcal{N}(\mu, \sigma)$. Standardizing the normal distribution means rescaling its standard deviation to unity, $\sigma = 1$.

The χ^2 -Distribution

Instead of a single random variable, what if we draw N standard normal rv's independently, $X_i \sim \mathcal{N}(0, 1)$, and form a new rv by taking the sum of their squares:

$$Y = \sum_{i=1}^N X_i^2 \quad (30)$$

what is $\rho_Y(y)$? To find it we need to resort to the joint probability distribution of both Y and all the X_i , and use Bayes' rule:

$$\begin{aligned} \rho_Y(y) &= \left[\prod_{i=1}^N \int_{-\infty}^{\infty} dx_i \right] \rho(y, x_1, \dots, x_N) \\ &= \left[\prod_{i=1}^N \int_{-\infty}^{\infty} dx_i \right] \rho(y|x_1, \dots, x_N) \rho_X(x_1) \cdots \rho_X(x_N) \\ &= \frac{1}{(2\pi)^{N/2}} \left[\prod_{i=1}^N \int_{-\infty}^{\infty} dx_i \right] \delta\left(y - \sum_i x_i^2\right) e^{-\frac{1}{2} \sum_i x_i^2} \end{aligned} \quad (31)$$

In the second line we used the independence of the X_i to factor their joint probability distribution. By examining the integral, we see that we are integrating over $\mathbb{R}^N$, and the delta function is projecting out all but the surface of a hypersphere of radius $\sqrt{y}$. Since the integrand is solely a function of the radius, the integral will evaluate to the surface area of this sphere times the value of the integrand at $r = \sqrt{y}$. Don't forget the extra $\frac{1}{2\sqrt{y}}$ factor coming from the delta function integration! The result is:

$$\rho_Y(y) = \frac{1}{2^{\frac{N}{2}} \Gamma(\frac{N}{2})} y^{\frac{N}{2}-1} e^{-\frac{1}{2}y} \quad (32)$$

We call this a χ^2 distribution with N degrees of freedom.

If instead of standardized random variables we had constructed the weighted sum of squares of nonstandard Gaussians:

$$Y = \frac{1}{N} \sum_{i=1}^N X_i^2 \quad (33)$$

$$\langle X_i \rangle = 0 \quad (34)$$

$$\langle X_i X_j \rangle = \sigma^2 \delta_{ij} \quad (35)$$

then the pdf for y will be:

$$\rho_Y(y|\sigma^2) = \frac{1}{\Gamma(\frac{N}{2})} \left(\frac{N}{2}\right)^{\frac{N}{2}} \frac{y^{\frac{N}{2}-1}}{\sigma^N} e^{-\frac{N}{2} \frac{y}{\sigma^2}} \quad (36)$$

This form of the χ^2 -distribution will be helpful later.

The Central Limit Theorem

We draw a standardized rv X from a probability density function (pdf), $\rho_X(x)$. Standardized means that $\langle X \rangle = 0$ and $\langle X^2 \rangle = 1$. We can always accomplish this by rescaling the rv:

$$X \rightarrow X' = \frac{X - \langle X \rangle}{\langle (X - \langle X \rangle)^2 \rangle^{\frac{1}{2}}} \quad (37)$$

Consider drawing independently N of these rv's, $\{X_i\}$ and defining a new rv

$$Y = \frac{1}{\sqrt{N}} \sum_{i=1}^N X_i \quad (38)$$

The question we wish to answer is what the pdf of this new variable, $\rho_Y(y)$, is in the limit $N \rightarrow \infty$. The joint probability density and Bayes' rule relate it to the pdf of X :

$$\begin{aligned} \rho_Y(y) &= \left(\prod_{i=1}^N \int dx_i \right) \rho(y, x_1, \dots, x_N) \\ &= \left(\prod_{i=1}^N \int dx_i \right) \rho(x_1, \dots, x_N) \rho(y | x_1, \dots, x_N) \\ &= \left(\prod_{i=1}^N \int dx_i \rho_X(x_i) \right) \delta \left(y - \frac{1}{\sqrt{N}} \sum_j x_j \right) \end{aligned} \quad (39)$$

where we have used the independence of the X_i 's to factor their joint pdf. The probability of a particular value of Y is the sum of the probabilities of all the different ways the X_i sum up to make that value. Rather than knowing this relationship, it would be nice to know the analytical form of ρ_Y . Spoiler Alert: it's a standardized Gaussian.

To prove this remarkable conclusion we will have to examine the characteristic functions of ρ_X and ρ_Y . The characteristic function can be thought of in two ways: as the expectation value of a plane wave, or as the Fourier transform of the pdf:

$$\begin{aligned} \psi_X(k) &= \int dx \rho_X(x) e^{ikx} \\ &= \langle e^{ikX} \rangle \\ &= \sqrt{2\pi} (\mathcal{F}\rho_X)(k) \end{aligned} \quad (40)$$

Both points of view will be useful; in particular that $\rho_X = \frac{1}{\sqrt{2\pi}} \mathcal{F}^{-1} \psi_X$. What is ψ_Y ?

$$\begin{aligned} \psi_Y(k) &= \int dy \rho_Y(y) e^{iky} \\ &= \int dy \int dx_1 \cdots \int dx_N \rho_X(x_1) \cdots \rho_X(x_N) \delta \left(y - \frac{1}{\sqrt{N}} (x_1 + \cdots + x_N) \right) e^{iky} \\ &= \left(\int dx \rho_X(x) e^{i \frac{k}{\sqrt{N}} x} \right)^N \\ &= \psi_X \left(\frac{k}{\sqrt{N}} \right)^N \end{aligned} \quad (41)$$

Ok, so that's really cool. Now if we put on our expectation value glasses for the characteristic function, and consider the large N behavior of the function:

$$\begin{aligned}
 \psi_Y(k) &= \lim_{N \rightarrow \infty} \langle e^{ikX/\sqrt{N}} \rangle^N \\
 &= \lim_{N \rightarrow \infty} \left\langle 1 + \frac{ikX}{\sqrt{N}} - \frac{k^2 X^2}{2N} + \dots \right\rangle^N \\
 &= \lim_{N \rightarrow \infty} \left(1 - \frac{k^2}{2N} + \dots \right)^N \\
 &= e^{-\frac{1}{2}k^2}
 \end{aligned} \tag{42}$$

Putting on our Fourier transform goggles, we recall that the (inverse) Fourier transform of a Gaussian yields another Gaussian:

$$\rho_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}y^2} \tag{43}$$

Voila! Y follows a standardized Gaussian. All hints of the underlying distribution, ρ_X , have vanished.

Our goal now will be to understand what happens when N is very large, but not infinite. We will see that the moments of ρ_X will play a crucial role in understanding the deviations of ρ_Y from a standardized Gaussian.

Edgeworth Expansion

The Edgeworth expansion is conceptually simple, but algebraically almost a Sisyphean task. We simply have to collect all the terms in the characteristic function in powers of $N^{-1/2}$, without throwing out higher order terms like we did in proving the CLT. Shall we begin, Clarrise?

Let's denote the m^{th} moment of X by $\langle X^m \rangle = \kappa_m$. Everything as before, except for the slight change in notation $\epsilon = N^{-1/2}$, the characteristic function for Y reads

$$\begin{aligned}
 \psi_Y(k) &= \left(1 - \frac{k^2}{2N} + \sum_{m=3}^{\infty} \frac{(ik)^m}{m!} \kappa_m \epsilon^m \right)^N \\
 &= \left(1 - \frac{k^2}{2N} \right)^N \left(1 + \sum_{\ell=0}^{\infty} \sum_{m=3}^{\infty} \frac{i^m k^{2\ell+m}}{2^\ell m!} \kappa_m \epsilon^{2\ell+m} \right)^N
 \end{aligned} \tag{44}$$

where the second sum came from a geometric series expansion. Yes, I know I'm mixing ϵ 's and N 's, and for good reason. In the $N \rightarrow \infty$ limit, the term on the outside of this expression is exactly $e^{-\frac{1}{2}k^2}$. With the awesome power of algebra, the binomial expansion, and a Taylor series, one can find the error of this equation:

$$\left(1 - \frac{k^2}{2N} \right)^N = e^{-\frac{1}{2}k^2} \left(1 - \frac{k^4}{8}\epsilon^2 + \left(\frac{k^8}{2^8} - \frac{k^6}{3 \cdot 2^3} \right) \epsilon^4 + \dots \right) \tag{45}$$

We're going to use this, as soon as we expand out the other piece in powers of ϵ . First off let's define

$$\alpha_j = \sum_{\ell=0}^{\lfloor \frac{j-3}{2} \rfloor} \frac{(ik)^j \kappa_{j-2\ell}}{(-2)^\ell (j-2\ell)!} \quad (46)$$

We list the first few below

$$\begin{aligned} \alpha_3 &= \frac{(ik)^3}{3!} \kappa_3 \\ \alpha_4 &= \frac{(ik)^4}{4!} \kappa_4 \\ \alpha_5 &= \frac{(ik)^5}{5!} \kappa_5 - \frac{(ik)^5}{2 \cdot 3!} \kappa_3 \end{aligned} \quad (47)$$

Applying the binomial expansion once more, we can now write the characteristic function as

$$\begin{aligned} \psi_Y(k) &= e^{-\frac{1}{2}k^2} \left(1 - \frac{k^4}{8}\epsilon^2 + \left(\frac{k^8}{2^8} - \frac{k^6}{3 \cdot 2^3} \right) \epsilon^4 + \dots \right) \left(1 + \sum_{j=3}^{\infty} \alpha_j \epsilon^j \right)^N \\ &= e^{-\frac{1}{2}k^2} \left(1 + \alpha_3 \epsilon + \left(\alpha_4 + \frac{1}{2} \alpha_3^2 - \frac{1}{8} k^4 \right) \epsilon^2 + \left(\frac{1}{6} \alpha_3^3 - 2 \alpha_3 + \alpha_3 \alpha_4 + \alpha_5 - \frac{1}{8} \alpha_3 k^4 \right) \epsilon^3 + \dots \right) \\ &= \left(1 + \frac{\kappa_3}{6} (ik)^3 \epsilon + \left(\frac{\kappa_4 - 3}{24} (ik)^4 + \frac{\kappa_3^2}{72} (ik)^6 \right) \epsilon^2 + \dots \right) e^{-\frac{1}{2}k^2} \end{aligned} \quad (48)$$

Maybe one day I'll write out the next two terms. Not today. We're almost there; the last step is to inverse Fourier transform, but we have all those pesky ik factors sticking around. They're not so bad if you remember this nice little identity

$$\mathcal{F}^{-1}[(ik)^m \rho] = (-1)^m \frac{\partial^m}{\partial x^m} \mathcal{F}^{-1} \rho \quad (49)$$

Using it to transform back to the pdf of Y we find

$$\rho_Y(y) = \left(1 - \frac{\kappa_3}{6\sqrt{N}} \frac{\partial^3}{\partial y^3} + \frac{\kappa_4 - 3}{24N} \frac{\partial^4}{\partial y^4} + \frac{\kappa_3^2}{72N} \frac{\partial^6}{\partial y^6} + \dots \right) \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}y^2} \quad (50)$$

We can make one final simplification if we recall the definition of the Hermite polynomials, which given that physicists are all obsessed with the simple harmonic oscillator you do, so the Edgeworth expansion reads

$$\rho_Y(y) = \left(1 + \frac{\kappa_3}{6\sqrt{N}} H_3(y) + \frac{\kappa_4 - 3}{24N} H_4(y) + \frac{\kappa_3^2}{72N} H_6(y) + \dots \right) \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}y^2} \quad (51)$$

As promised, the skewness and kurtosis of the underlying distribution contribute to the first and second order terms of the Edgeworth expansion. We will primarily concern ourselves with the first order correction. The appearance of the third order moment here will have an analogy when we start examining the CMB. There we will find the three point correlation function, or in Fourier space, the bispectrum.